

Francesco Sala

Mechanical Engineering M.Sc. Student
Motorsport | Structural Design | Composite Materials

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PROFILE

Curious and driven engineer with four years of experience in the Polimi Motorcycle Factory, where I developed hands-on expertise in the design and development of racing components. I thrive in fast-paced, collaborative environments that require precision, continuous improvement, and practical problem-solving. Beyond engineering, I am strongly motivated by innovation, value creation, and the development of new ideas in technology-driven industries.

EDUCATION

Master of Science <i>Computational Mechanical Design (CM3 track)</i> Politecnico di Milano	Sep. 2024 – Present Milan, Italy
Bachelor of Science <i>Mechanical Engineering (Propaedeutics track)</i> Politecnico di Milano	Sep. 2021 – July 2024 Milan, Italy

WORK EXPERIENCE

Mechanical Design Intern, Motorsport YCOM	Mar. 2026 – Present Collecchio (PR), Italy
- Designed and optimized high-performance composite, metallic and 3D printed motorsport components using CATIA V5. - Developed comprehensive technical drawings, documentation, and geometric dimensioning and tolerancing layouts to support manufacturing teams.	
Structural Engineer Polimi Motorcycle Factory	Nov. 2022 – Present Milan, Italy
- Engineered primary structural motorcycle components, including carbon fiber chassis and swingarm, following the design from initial concept to track validation. - Analyzed complex mechanical assemblies in Abaqus and 3DEXPERIENCE, conducting FEA simulations to optimize composite layup lamination, stiffness, and structural integrity. - Supervised hands-on carbon fiber lamination, structural bonding, vacuum-bagging, and curing processes, ensuring strict compliance with tight racing schedules. - Collaborated in a fast-paced, multidisciplinary engineering team, solving complex dynamic integration problems under demanding race deadlines.	

Others jobs: Waiter, Bartender and Cashier Poormanger, Antica Focacceria San Francesco	Nov. 2023 – Mar 2026 Milan, Italy
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SKILLS

Languages: Italian (Native), English (C1 – TOEIC 955/990).
CAD Design Tools: 3DEXPERIENCE, Solid Edge, Autodesk Inventor, SolidWorks, CATIA V5, OnShape.
Programming: Python, MATLAB.
Simulation Engineering Tools: Simulink, Simscape, Abaqus, OpenFOAM, Altair.
Web Development Version Control: HTML5, CSS3 (base), Git, GitHub.
AI-Assisted Development: OpenAI Codex, ChatGPT, Claude, Google Gemini; prompt engineering and debugging.
Digital Documentation Tools: Microsoft Office Suite, LaTeX, Canva, Figma.

PROJECTS AND RESEARCH

AI-Assisted CAD Automation for CATIA V5 <i>Python, OpenAI Codex</i> Personal project	SUMMER 2026
Design, validation and MVP launch of a carpooling platform <i>User Interviews, MVP Test</i> Politecnico di Milano	SPRING 2026
Design and analysis of a carbon fiber McPherson suspension <i>Abaqus, Catia</i> Politecnico di Milano	WINTER 2025
Design and production of a TPU capsule via additive manufacturing <i>Bambulab, Catia</i> Politecnico di Milano	WINTER 2025
Design of a Metal load-bearing arm via additive manufacturing <i>Altair, Catia</i> Politecnico di Milano	WINTER 2025
Design and production of a carbon fiber Chassis <i>Surface Design and Composite Lay-Up CAD</i> Polimi Motorcycle Factory	FALL 2025
LLM-Assisted Process Planning for CNC Machining <i>Python, CadQuery, OpenAI API</i> Politecnico di Milano	SPRING 2025
Bayesian Optimization of a PID Controller for a 7-Axis Robot <i>MATLAB, B.O.</i> Politecnico di Milano	SPRING 2025
Numerical Simulation of Internal Flow Through a Converging Nozzle <i>OpenFoam</i> Politecnico di Milano	SPRING 2025
Turbulent Flow Around a NACA 4412 Airfoil <i>OpenFoam, MATLAB</i> Politecnico di Milano	SPRING 2025
Dynamic Characterization and Experimental Modal Analysis of M.S. <i>MATLAB</i> Politecnico di Milano	SPRING 2025
Design and production of a carbon fiber Swingarm <i>Surface Design and Composite Lay-Up</i> Polimi Motorcycle Factory	SPRING 2023

SPORT EXPERIENCE

CUS Pro Patria Milano <i>Sprinter 100, 200 400 mt</i> Milan, Italy	Jan. 2025 – Present
CUS Parma <i>Sprinter 100, 200 400 mt</i> Parma, Italy	Dec. 2010 – Sep. 2021